2022 SSMO Relay Round 1

SMO Team

RR 1 Part 1: Suppose a, b, c are distinct digits where $a \neq 0$ such that $\left(\overline{abc}\right)^2 = \overline{bad00}$ where $d = a + b$. Find $a + 2b$.

RR 1 Part 2: Let $T = TNYWR$. Suppose lines ℓ and m have equations $y = (2+\sqrt{3})x+16$ and $y = \frac{\sqrt{3}}{3}x + 20$, respectively. Let A be a point on ℓ such that the shortest distance from A to m is T. Let O be the point on m such that $\overline{AO} \perp m$, and let P be the point on ℓ such that $PO \perp \ell$. Find PO^2.

RR 1 Part 3: Let $T = TNYWR$. Consider triangle ABC with $AB = T$, $AC = 100$, and $\angle ABC = 36°$. Find the product of all possible values of BC.

2022 SSMO Relay Round 2

SMO Team

RR 2 Part 1: Let P be a randomly selected point on a circle, and let A be a randomly selected point inside the same circle. A dilation centered at P with a scale factor of 2 sends A to A'. The probability that PA' is less than the length of the diameter of the circle can be expressed as $\frac{a\pi-b\sqrt{c}}{d\pi}$, where a, b, c, and d are positive integers, c is not divisible by the square of any prime, and $\gcd(a,b,d)=1$. Find $a+b+c+d$.

RR 2 Part 2: Let $T = TNYWR$. Suppose that the monic quadratic $f(x)$ is tangent to the function $g(x) = |x+2| - T$ at two points. The value of $|f(1)|$ can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

RR 2 Part 3: Let $T = TNYWR$. Suppose that

$$x + y = \left\lfloor \sqrt{T} \right\rfloor \quad \text{and}$$
$$x^5 + y^5 = 15.$$

The absolute difference between the two possible values of xy can be expressed as $\frac{a\sqrt{b}}{c}$, where a, b, and c are positive integers, a and c are relatively prime, and b is not divisible by the square of any prime. Find $a+b+c$.

2022 SSMO Relay Round 3

SMO Team

RR 3 Part 1: Let $f : \mathbb{Z} \to \mathbb{Z}$ be a function such that $f(0) = 0$ and $f(|x^2-4|) = 0$ whenever $f(x) = 0$. Suppose further that $|f(x+1) - f(x)| = 1$ for all $x \in \mathbb{Z}$. Let N be the number of possible sequences $\{f(1), f(2), \ldots, f(21)\}$. Find the remainder when N is divided by 1000.

RR 3 Part 2: Let $T = TNYWR$. In cyclic quadrilateral $ABCD$, $\angle BAD = 60^\circ$ and $BC = CD = T$. Suppose that AB is a positive integer. Find the median of all possible values of $2AB$.

RR 3 Part 3: Let $T = TNYWR$. Suppose $f(x)$ is a polynomial of degree 10, such that $f(i) = i$ for all $i = 1, 2, \ldots, 10$ and $f(11) = T$. Find the remainder when $f(13)$ is divided by 1000.

2022 SSMO Relay Round 4

SMO Team

RR 4 Part 1: On each day, a robot finds a new organism with probability 0.7, an already discovered organism with probability 0.2, and nothing with probability 0.1. If a new organism is found, the robot correctly identifies it as new with probability 0.9, and if a discovered organism is found, the robot correctly identifies it as discovered with probability 0.75. The expected number of days until the robot reports that it has found a new organism can be expressed as $\frac{m}{n}$, where m and n are relatively prime positive integers. Find $m+n$.

RR 4 Part 2: Let $T = TNYWR$. The roots of $f(x) = x^3 + 5x + 8$ are r_1, r_2, r_3. Let $g_n(x)$ be a monic cubic polynomial with roots $r_1 + n, r_2 + n, r_3 + n$. Find the remainder when

$$\sum_{n=1}^{T} (-1)^n g_n(5)$$

is divided by 1000.

RR 4 Part 3: Let $T = TNYWR$. Suppose $f : \mathbb{Z} \to \mathbb{Z}$ satisfies $f(1) = 1$, $f(2) = 12$, and

$$f(n+2) = 12f(n+1) - 20f(n)$$

for all positive integers n. Find the remainder when $f(T)$ is divided by 1000.

2022 SSMO Relay Round 5

SMO Team

RR 5 Part 1: Consider an 8×8 chessboard with a knight in one of the center squares. The knight may move in an L-shaped fashion, going two squares in one direction and one square in a perpendicular direction, but cannot go outside the chessboard. How many squares can the knight reach in exactly two moves?

RR 5 Part 2: Let $T = TNYWR$. Consider sequence $S = \{a_1, a_2, \ldots, a_{2022}\}$ of positive integers such that $a_1 \leq a_2 \leq \cdots \leq a_{2022}$ and $\text{lcm}(a_1, a_2, \ldots, a_{2022}) = 70T$. Assume $\gcd(a_i, a_j) = 1$ for all $1 \leq i < j \leq 2022$. Find the number of possible sequences S.

RR 5 Part 3: Let $T = TNYWR$. Define $\text{cis}(x) = \cos(x) + i \sin(x)$ and $a_k = \text{cis}\left(\frac{k\pi}{T+1}\right)$. The value of

$$\sum_{k=1}^{T+1} \frac{|a_{2k} + a_{2k+2} - a_{2k-1+T}|}{|a_{2k+1} - (a_{2k} + a_{2k+2})|}$$

can be expressed as $a + b\cos(\frac{\pi}{c})$, where a, b, and c are positive integers. Find $a + b + c$.